

Advanced Trigonometry

Self-Test 1

Allow 25 minutes to complete this test.

1. Find the length of side c in $\triangle ABC$ if $A = \frac{\pi}{4}$, $B = \frac{\pi}{6}$ and $b = 4$.
2. Find the size of the smallest angle in $\triangle ABC$ if the side lengths are $a = 2$, $b = 4$ and $c = 5$.
3. A hillside slopes at 20° to the horizontal. A 10-metre pole stands vertically upright about halfway up the hill. How long a rope is needed to reach from the top of the pole to a point 20 metres downhill from the base of the pole?
4. Graph the function $y = f(x) = 2 \sin 3x$ noting its amplitude and period.
5. Graph the function $y = f(x) = \frac{1}{2} \tan x$.
6. If $\tan x = \frac{4}{3}$ and $\pi \leq x \leq \frac{3\pi}{2}$, evaluate $\cos x$ using trigonometric identities.
7. Simplify the expression $\frac{1}{1 + \cos x} + \frac{1}{1 - \cos x}$.
8. True or false: $\sin\left(x + \frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}(\sin x + \cos x)$.
9. True or false: $\cos(\pi - x) = \sin\left(\frac{\pi}{2} - x\right)$.
10. Show that $\frac{\sin 2x}{\sin x} - \frac{\cos 2x}{\cos x} = \sec x$.